

Median (अंशक):

Ungrouped:

$$\text{median} = \begin{cases} \frac{n+1}{2} \text{th, if } n \text{ is odd} \\ \frac{\frac{n}{2} + (\frac{n}{2} + 1)}{2} \text{th observation, if } n \text{ is even.} \end{cases}$$

* 82, 79, 60, 76, 63, 81, 68, 74, ~~60~~ 75;
find median.

⇒ In sequence: ⁶⁰60, ⁶⁸63, ⁶74, ⁶75, ⁶76, ⁶79, 81, 82.

$$\therefore n = 9$$

$$\therefore \text{median} = \frac{n+1}{2} \text{th} \\ = \frac{9+1}{2} \text{th} \\ = 5 \text{th} = 75$$

$$\therefore n = 10$$

$$\therefore \text{median} = \frac{\frac{10}{2} + (\frac{10}{2} + 1)}{2} \text{th}$$

$$= \frac{5 \text{th} + 6 \text{th}}{2} \\ = \frac{74 + 75}{2} = 74.5, \text{ Ans}$$

L_1 = lower limit where mode occurs.

f_m = frequency of mode class.

f_1 = preceding frequency of mode class

f_2 = post

C = difference between limits of mode class.

question:

$$\bullet \text{ mean} = a + \frac{h}{N} \sum f_i d_i$$

$$\bullet \text{ median} = L + \frac{(n/2 - c)h}{f}$$

$$\bullet \text{ mode} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times c$$

Ans:

Age Group	No of patients (f_i)	C. frequency	avg, x_i	d_i	$f_i d_i$
25-30	4	4	27.5	-4	-16
30-35	3	7	32.5	-3	-9
35-40	2	9	37.5	-2	-4
40-45	8	12	42.5	-1	-3
45-50	4	16	47.5 $\leftarrow a$	0	0
50-55	8	24	52.5	1	8
55-60	6	30 = N	57.5	2	12

$$\sum f_i d_i = -12$$

Here, $a \rightarrow 47.5$

$h \rightarrow 5$

$N \rightarrow 30$

$\sum f_i d_i \rightarrow -12$

$$\begin{aligned} \therefore \text{mean, } \bar{x} &= a + \frac{h}{N} \sum f_i d_i \\ &= 47.5 + \frac{5}{30} \cdot (-12) \\ &= 47.5 - 2 = 45.5 \end{aligned}$$

$$L = 45$$

$$n_2 = 30/2 = 15$$

$$c = 12$$

$$h = 5$$

$$f = 16$$

$$\therefore \text{median} = L + \frac{(n_2 - c) \cdot h}{f}$$

$$= 45 + \frac{(15 - 12) \cdot 5}{16}$$

$$= 45 + 0.9375$$

$$= 45.9375 \text{ (Ans)}$$

∴ Here,

$$L = 50$$

$$c = 5$$

$$f_m = 8$$

$$f_1 = 4$$

$$f_2 = 6$$

$$\text{Mode} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times c$$

$$= 50 + \frac{8 - 4}{2 \cdot 8 - 4 - 6} \times 5$$

$$= 50 + 3.33$$

$$= 53.33 \text{ (Ans)}$$

HW: Sheet 4
Chapter 5
2.7 math.

• Group data:

$$\text{Median} = L + \frac{\left(\frac{n}{2} - c\right)h}{f}$$

L = Lower observation of median class.

n = total . . .

c = preceding cumulative frequency of median class.

h = difference of limits of median class

f = frequency of median class.

☐ Mode (अंश)

• Ungroup: निम्नलिखित सूत्रों से मोड निकाला जाता है

• Group data:

$$\text{Mode} = L_1 + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times C$$

C = Cumulative frequency - मूल्य f_1